

Josh Hall

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SUMMARY

Software engineering student at Cal Poly in the **blended B.S./M.S. program**, with backend and systems experience from an internship at **Hewlett Packard Enterprise Storage**. **Co-author** of a **VR research paper** submitted to ACM CHI PLAY 2026. Special interest in Deep Learning, Machine Learning, Computer Vision, and AI. Experienced across **Python backend services, deep learning in PyTorch, full stack applications, and Unity and VR development**.

EDUCATION

California Polytechnic State University (Cal Poly) - San Luis Obispo, CA

Sept 2022 – Jun 2027

B.S. Software Engineering + M.S. Computer Science (Blended Program) | GPA 3.7, Dean's List

TECHNICAL SKILLS

- **Languages:** Python, C#, C, C++, Java, Kotlin, TypeScript, JavaScript, SQL, Bash
- **Frameworks & Libraries:** PyTorch, scikit-learn, React, Node.js, Express, FastAPI, gRPC, Protocol Buffers
- **Data & Cloud:** PostgreSQL, MongoDB, Supabase, AWS, Azure, Kubernetes, Docker, Render
- **Tools & Engines:** Git, Linux, pytest, Unity, Unreal Engine 5, Godot, WebGL, Android Studio

EXPERIENCE

Software Engineering Intern (Python, C++, OOP) - Hewlett Packard Enterprise, Storage

June 2026 – Aug 2026

- Rewrote a **2,700-line legacy C++ diagnostics CLI** for an enterprise storage platform as an **object oriented Python tool** and client library, cutting the lines of code to edit the same command to **190 lines (93% less code)** while adding functionality.
- Designed a resource-verb command grammar modeled on industry-standard cloud CLIs, **shortening typical invocations by 82%** and adding machine-readable JSON and YAML output alongside human-readable text.
- Built **90 automated tests to 100% coverage** for a tool that previously had none, injecting a mocked transport layer so the full suite runs in **under a second** with no hardware or cluster required.
- Implemented **role-based access control** across four privilege tiers from a single declarative policy table, enforced at argument-parse time and designed to **fail closed**, with a test that turns an unclassified command into a build failure.
- Engineered a **gRPC and Protocol Buffers** client against an internal RPC framework, including a generic call envelope, connection keepalive, and selective retry with backoff that distinguishes transient from permanent failures.

Undergraduate Researcher - Ocean Sight One (VR, Unity, C#) - Cal Poly

Aug 2025 – Present

- Built **mini-games** in Unity implementing **pathfinding AI** for fish and a parameterized roaming state with adjustable height and width bounds.
- Animated barracuda and shark models in **Unity's animation editor** to give the predators lifelike motion to increase immersion.
- Ran **user testing with 50+ participants** across the year, iterating mini-game mechanics and onboarding on the feedback.
- **Co-authored research paper** "*Ocean Sight One: Making Offshore Oil Platform Ecology Legible Through Play*" submitted to ACM CHI PLAY 2026, on communicating contested environmental science through embodied VR gameplay.

Software Developer - Vet Scientia: Veterinary Anesthesia Training Platform (Unity, C#, React) - Cal Poly

Aug 2025 – June 2026

- Developed the Unity simulation - half of a veterinary anesthesia training platform for **university vet programs and clinics**, paired with a React, Express, and MongoDB dashboard for course and assignment management.
- **Partnered with the Pulse Physiology Engine team** to secure special access for veterinary use, then integrated the engine to drive animal vital signs in real time and log them for post-lesson performance graphs.
- Assembled the operating-room scene and 3D animal models, correcting materials, textures, and lighting so the environment reads accurately to students.
- Wrote core simulation scripts for **lesson loading and JSON save/restore**, surgeon notification popups, and the **WebGL bridge** that pulls per-user save files from the dashboard and returns results for grading.

PROJECTS

Artist Classification from Paintings (PyTorch, ResNet-18, WikiArt)

Jan 2026 – March 2026

- Built a **20-class painting-to-artist classifier** in PyTorch on a balanced 10,000-image WikiArt subset with a team of 5, reaching **84.2% top-1** and **95.4% top-5** test accuracy (**0.836 macro F1**).
- Replaced a from-scratch CNN with a pretrained **ResNet-18** backbone and a **two-phase fine-tuning schedule**, training the classification head against a frozen backbone before unfreezing deeper layers at a 10x lower learning rate.
- Designed a painting-specific augmentation pipeline and trained with AdamW, cosine annealing, label smoothing, and early stopping, using confusion matrices to diagnose confusions between stylistically similar artists.

Potion Project (SQL, PostgreSQL, Supabase, Render, Python)

March 2025 - June 2025

- Built and deployed a **production-ready FastAPI Python backend** integrated with **PostgreSQL** (via **Supabase**) and hosted on **Render**, implementing database migrations with Alembic and enforcing data integrity with **SQL** constraints.
- Designed and normalized relational schemas to support **dynamic inventory management**, custom potion recipes, and order/cart systems with proper foreign key relationships.
- Implemented advanced backend patterns, including event-sourced **ledgerization** and **idempotent API endpoints** to ensure fault tolerance, consistency, and accurate financial tracking.
- Developed analytics and search functionality by adding instrumentation, writing SQL queries for data-driven strategies, and creating efficient search/filter/pagination endpoints for **order management**.
- Placed **top 4% of ~200** students in a course wide simulation competition scored on in-game revenue.